GPU BUILDER · BUILD GUIDE

My first graphics card

A step-by-step assembly reference for the GP–01 teaching card: 22 steps and 47 reference designators, from the bare PCB to the power cable.

GENERATED	ASSEMBLIES	KNOWLEDGE CHECKS	MODE
October 24, 2026	0 of 47 placed	0 of 22 correct	Guided placement

Before you start

- Work at an ESD-safe bench and ground yourself before touching the board.
- Never assemble or re-seat a card with the power supply connected.
- Check every connector's keying and pin order against its cable before applying power.
- Follow the manufacturer's mounting hardware, torque and thermal-interface instructions for any part you buy.

This guide is a teaching reference. Geometry, ratings and phase counts are illustrative; no schematic, netlist, Gerber, pin mapping or manufacturable design is included, and the checks do not replace electrical, thermal or mechanical verification.

Parts to source

Every reference on the board, with what it is and where a builder realistically gets it: 23 hardware · 17 distributor · 2 fabrication · 5 integrated.

Silicon sold only to board manufacturers is marked integrated. Hardware and distributor parts are ordinary purchases; a salvage donor card covers the integrated items plus the fabricated PCB.

REF.	PART	QTY	SOURCE	WHAT TO BUY
PCB1	Multilayer PCB	1	FABRICATION	Order from a PCB fabricator: multilayer stackup, controlled impedance, ENIG finish.
J1	PCIe edge interface	1	FABRICATION	A gold-fingered board edge, not a separate part. Specify the card-edge bevel in fabrication.
J2	Auxiliary power input	1	DISTRIBUTOR	Auxiliary power connector rated for the rail; keying and pin count per the connector standard you chose.
U6	PWM controller	1	DISTRIBUTOR	Multiphase PWM controller; match the phase count and control interface.
U8	Integrated power stage 1	1	DISTRIBUTOR	Integrated power stage rated above the phase current, with thermal margin.
U9	Integrated power stage 2	1	DISTRIBUTOR	Integrated power stage rated above the phase current, with thermal margin.
U10	Integrated power stage 3	1	DISTRIBUTOR	Integrated power stage rated above the phase current, with thermal margin.
U11	Integrated power stage 4	1	DISTRIBUTOR	Integrated power stage rated above the phase current, with thermal margin.
L1	Phase inductor 1	1	DISTRIBUTOR	Saturation current above the phase peak; choose DCR and height to suit the board.
L2	Phase inductor 2	1	DISTRIBUTOR	Saturation current above the phase peak; choose DCR and height to suit the board.
L3	Phase inductor 3	1	DISTRIBUTOR	Saturation current above the phase peak; choose DCR and height to suit the board.
L4	Phase inductor 4	1	DISTRIBUTOR	Saturation current above the phase peak; choose DCR and height to suit the board.
C1	Bulk capacitor 1	1	DISTRIBUTOR	Voltage rating, capacitance and ESR for the output network; polymer or ceramic.
C2	Bulk capacitor 2	1	DISTRIBUTOR	Voltage rating, capacitance and ESR for the output network; polymer or ceramic.
U1	GPU package	1	INTEGRATED	Not sold as a loose part. A packaged GPU comes on an assembled board.
U2	GDDR6 memory 1	1	INTEGRATED	GDDR packages are sold to board makers, not as individual parts. Source a donor board.
U3	GDDR6 memory 2	1	INTEGRATED	GDDR packages are sold to board makers, not as individual parts. Source a donor board.
U4	GDDR6 memory 3	1	INTEGRATED	GDDR packages are sold to board makers, not as individual parts. Source a donor board.
U5	GDDR6 memory 4	1	INTEGRATED	GDDR packages are sold to board makers, not as individual parts. Source a donor board.
U7	SPI firmware flash	1	DISTRIBUTOR	SPI NOR flash in the package and voltage the board supports.
Y1	Reference oscillator	1	DISTRIBUTOR	Oscillator at the reference frequency with the stability the design needs.
C3–C10	Decoupling capacitor bank	8	DISTRIBUTOR	Ceramic capacitors by package, voltage and dielectric; the board uses eight here.

REF.	PART	QTY	SOURCE	WHAT TO BUY
J3	Display output connector	1	DISTRIBUTOR	Display connector matching the interface and the bracket cut-outs.
J4	Fan header	1	DISTRIBUTOR	4-pin PWM header; check the pin order against the fan lead.
H1	Corner standoff 1	1	HARDWARE	M3 male-female standoffs; height must match the cooler and backplate hardware.
H2	Corner standoff 2	1	HARDWARE	M3 male-female standoffs; height must match the cooler and backplate hardware.
H3	Corner standoff 3	1	HARDWARE	M3 male-female standoffs; height must match the cooler and backplate hardware.
H4	Corner standoff 4	1	HARDWARE	M3 male-female standoffs; height must match the cooler and backplate hardware.
TP1	Memory thermal pad 1	1	HARDWARE	Thermal pad in the thickness that matches the gap; 1.0–1.5 mm is common on cards.
TP2	Memory thermal pad 2	1	HARDWARE	Thermal pad in the thickness that matches the gap; 1.0–1.5 mm is common on cards.
TP3	Memory thermal pad 3	1	HARDWARE	Thermal pad in the thickness that matches the gap; 1.0–1.5 mm is common on cards.
TP4	Memory thermal pad 4	1	HARDWARE	Thermal pad in the thickness that matches the gap; 1.0–1.5 mm is common on cards.
TP5	Power-stage thermal pad	1	HARDWARE	Thermal pad in the thickness that matches the gap; 1.0–1.5 mm is common on cards.
TIM1	Thermal paste layer	1	HARDWARE	Non-conductive thermal paste; one small tube covers several mounts.
HS1	Heatsink assembly	1	HARDWARE	Salvage a fin stack or have one machined; the mounting holes must match the board.
FAN1	Cooler fan 1	1	HARDWARE	PWM fan sized to the shroud, with the same voltage and connector as the header.
FAN2	Cooler fan 2	1	HARDWARE	PWM fan sized to the shroud, with the same voltage and connector as the header.
FAN3	Cooler fan 3	1	HARDWARE	PWM fan sized to the shroud, with the same voltage and connector as the header.
BP1	Backplate	1	HARDWARE	Sheet aluminium or steel, or a salvage part; drill to the standoff pattern.
H5	Spring screw 1	1	HARDWARE	Spring-loaded cooler screws in the standoff thread; buy a matched set.
H6	Spring screw 2	1	HARDWARE	Spring-loaded cooler screws in the standoff thread; buy a matched set.
H7	Spring screw 3	1	HARDWARE	Spring-loaded cooler screws in the standoff thread; buy a matched set.
H8	Spring screw 4	1	HARDWARE	Spring-loaded cooler screws in the standoff thread; buy a matched set.
BR1	PCIe bracket	1	HARDWARE	Standard PCI bracket; check slot spacing and port openings before buying.
H9	Bracket screw 1	1	HARDWARE	Bracket screws in the board thread; check the length so they do not protrude into the card.
H10	Bracket screw 2	1	HARDWARE	Bracket screws in the board thread; check the length so they do not protrude into the card.
CB1	Auxiliary power cable	1	HARDWARE	Power-supply cable with the right connector at each end; verify the pinout.

Build steps

Work in order. Each step states what the part does, where it goes, what to watch for and a question to check your understanding.

01 Start with the substrate

To place

FOUNDATION · PCB1

Place PCB1. Every connection needs a home. Place the PCB to establish the mechanical foundation and the routing area for your graphics card.

What it does	A PCB supports the components. Copper traces, planes and plated vias connect circuits across its layers; the dielectric separates conductors.
Where it goes	The suggested outline reserves room for a host connector, display bracket and cooling hardware. The displayed geometry is a teaching layout, not a mechanical drawing.
Watch for	Real stackups are chosen for impedance, return paths, power distribution and manufacturability. A layer count alone does not establish signal quality.

Check yourself: Why place a ground reference near a high-speed signal layer? **Answer: To provide a controlled return path**

Reference: TI · High-speed PCB layout guidance

02 Connect to the host

To place

INTERFACES · J1

Place J1. Add the edge interface that connects the card to its host computer.

What it does	PCIe carries commands and data between the host and the graphics card. Its differential lanes are distinct from the GPU's local graphics-memory interface.
Where it goes	The connector sits on the board edge so it can mate with a motherboard slot. Actual contact dimensions and lane routing follow the applicable PCIe specifications.
Watch for	Lane count, link generation and protocol overhead determine host-link behavior. This model does not assign or validate pin mappings.

Check yourself: Does the PCIe link replace the GPU's local memory bus? **Answer: No, they connect different endpoints**

Reference: NVIDIA · GPU architecture fundamentals

03 Bring power onto the board

To place

POWER · J2

Place J2. Place the auxiliary DC input before assembling the core power converter.

What it does	The input connector brings external supply power onto the card. The board distributes that input to conversion stages and other rails.
Where it goes	An edge location leaves cable access and room for a protected input path. Connector selection, current rating and cable clearance require separate design work.
Watch for	The model uses a generic auxiliary connector. It does not assign a 12V-2x6 or 8-pin pinout, current limit, fuse rating or safe operating power.

Check yourself: Why not apply the auxiliary input directly to the GPU core? **Answer: A GPU core needs a regulated lower voltage**

Reference: TI · Multiphase processor power

04 Coordinate the power phases

To place

POWER · U6

Place U6. Install U6, the controller for the four-phase teaching supply.

What it does	A multiphase controller regulates a rail by coordinating switch timing and feedback. Phase interleaving can distribute current and reduce combined ripple.
Where it goes	Control and feedback connections must be planned around power stages and the regulated load, while avoiding noisy switching nodes.
Watch for	More phases do not guarantee better performance. Compensation, phase balance, sensing, operating modes and layout still matter.

Check yourself: What does phase interleaving change? **Answer: The timing of switching among phases**

Reference: TI · Multiphase processor power

05 Add the switching power stages

To place

POWER · U8 · U9 · U10 · U11

Place U8, U9, U10 and U11. Place U8 through U11, one integrated power stage for each phase.

What it does	Each stage combines a gate driver and power switches. It generates a switched waveform that feeds a phase inductor.
Where it goes	The suggested row keeps the power path compact. Switching-node area, heat flow and high-current return paths need careful physical design.
Watch for	These are integrated ICs, so their identifiers use U. Discrete MOSFETs would conventionally use Q designators. Four packages here do not define a current rating.

Check yourself: Which connection is especially sensitive to switching noise? **Answer: The switching node between stage and inductor**

Reference: TI · Multiphase processor power

06 Smooth each phase current

To place

POWER · L1 · L2 · L3 · L4

Place L1, L2, L3 and L4. Place L1 through L4 next to their corresponding switching stages.

What it does	The inductors store energy magnetically. Together with the output network, they turn switching pulses into a controlled DC supply.
Where it goes	Keep each stage-to-inductor connection compact and provide a suitable high-current path toward the load.
Watch for	Inductance, saturation current, DCR and core loss affect behavior. You cannot select a real inductor from the package shape alone.

Check yourself: Which inductor limit matters during a high-current transient? **Answer: Its saturation current**

Reference: TI · Multiphase processor power

07 Support load transients

To place

POWER · C1 · C2

Place C1 and C2. Add C1 and C2, the visible bulk capacitors in the teaching model.

What it does	Capacitance supplies and absorbs charge as the load changes. The full output network works with the regulator's control loop.
Where it goes	A real design distributes capacitance by impedance and current path. Two illustrated capacitors are not a complete GPU output network.
Watch for	Capacitance value, ESR, ESL, bias dependence and loop stability all matter. Build checks here do not analyze the regulator.

Check yourself: Is adding more capacitance always sufficient to fix a regulator? **Answer: No, stability and parasitics also matter**

Reference: TI · Multiphase processor power

08 Place the GPU package

To place

COMPUTE · U1

Place U1. Install U1. This is the packaged processor mounted to the PCB, not a bare silicon die placed directly on board traces.

What it does	The GPU executes parallel workloads. Its package carries signals and power between the silicon and the board through a ball-grid array.
Where it goes	The central position leaves space for nearby memory, power delivery and the cooler. Package escape routing is omitted from this model.
Watch for	Open the "Inside the GPU" view to distinguish functional silicon blocks from the discrete packages on the PCB.

Check yourself: Which item belongs inside the GPU silicon? **Answer: A streaming multiprocessor**

Reference: NVIDIA · GPU architecture fundamentals

09 Build the local memory interface

To place

MEMORY · U2 · U3 · U4 · U5

Place U2, U3, U4 and U5. Place the four graphics-memory packages, U2 through U5, around the processor.

What it does	GDDR holds working data such as textures, frame buffers and compute buffers. It exchanges data with the GPU's memory controllers.
Where it goes	Memory packages sit near the GPU to support short, controlled interconnects. Real topology and length matching depend on the devices and interface specification.
Watch for	Training parameters: each package is 2 GB with an independent x32 interface at 16 Gb/s per pin. Four packages illustrate 8 GB, 128 bits and 256 GB/s theoretical aggregate bandwidth. Compatibility with any real GPU is not implied.

Check yourself: For a 128-bit bus at 16 Gb/s per pin, what is the theoretical bandwidth? **Answer: 256 GB/s**

Reference: Micron · Graphics memory

10 Store board initialization data

To place

SUPPORT · U7

Place U7. Install U7, a small nonvolatile SPI flash device.

What it does	A firmware flash can store graphics-board initialization code and configuration data. Its contents persist without board power.
Where it goes	Keep its interface and programming access in mind. The teaching package is generic and has no downloadable vendor firmware.
Watch for	This device is not working graphics memory. Real firmware is coupled to the board and GPU implementation.

Check yourself: What distinguishes SPI flash from graphics DRAM? **Answer: Flash retains its contents without power**

Reference: Winbond · Code-storage flash memory

11 Establish a timing reference

To place

SUPPORT · Y1

Place Y1. Add Y1, the board's illustrative reference oscillator.

What it does	A reference oscillator supplies a stable periodic signal. Clock-generation circuits derive operating clocks from reference sources.
Where it goes	The location is illustrative. Real reference routing, loading, jitter and clock-domain requirements are device-specific.
Watch for	A reference oscillator is not the same as the GPU's final core frequency. PLLs and power-management controls may generate and adjust operating clocks.

Check yourself: Must the reference oscillator run at the final core clock frequency? **Answer: No, clock generation can derive other frequencies**

Reference: SiTime · Clock-generation reference example

12 Control local supply impedance

To place

SUPPORT · C3–C10

Place C3–C10. Place the C3–C10 capacitor bank near the processor. Eight parts are represented as one grouped assembly.

What it does	Local bypass capacitance helps manage supply impedance and fast load-current changes near an IC.
Where it goes	A short loop between power, capacitor and return is important. Many real decouplers are on the back side or within the package.
Watch for	This board view omits the complete power-distribution network and parasitic models. The visible bank does not represent all required capacitors.

Check yourself: Why can a distant bypass capacitor be less effective at high frequency? **Answer: Interconnect inductance adds impedance**

Reference: TI · High-speed PCB layout guidance

13 Provide a display connection

To place

INTERFACES · J3

Place J3. Fit J3 on the bracket edge, where a display cable can reach it.

What it does	A display output interface sends a processed display stream to a monitor through an appropriate transmitter and connector.
Where it goes	Keep connector access, signal integrity, ESD protection and mechanical support in view. This generic drawing does not specify HDMI or DisplayPort pinouts.
Watch for	A real connector does not directly expose internal GPU compute lanes. Protocol, PHY and protection details are outside this placement model.

Check yourself: What else must be considered beyond a connector's physical position? **Answer: Signal integrity and ESD protection**

Reference: NVIDIA · GPU architecture fundamentals

14 Connect the cooling control

To place

COOLING · J4

Place J4. Place J4 so the cooler's fan lead has a clear connection point.

What it does	A fan header may provide power, a speed-sense signal and a control signal, depending on its design.
Where it goes	Leave access for the cable and avoid interference with the cooler. Pin counts, voltage and control details vary.
Watch for	The diagram is not a fan pinout. Actual polarity, ratings and control standards must come from the selected devices.

Check yourself: Is a fan's speed-sense output the same as its power input? **Answer: No, they serve different roles**

Reference: Noctua · PWM fan specifications

15 Mount the standoffs

To place

ASSEMBLY · H1 · H2 · H3 · H4

Place H1, H2, H3 and H4. Before the cooler goes on, give it something to clamp against. Place H1 through H4, one in each corner hole.

What it does	Standoffs are threaded spacers. They hold a fixed gap between the board and the backplate or cooler bracket, and they carry the clamping load of the cooler screws so the PCB is not bent.
Where it goes	H1–H4 sit in the board's mounting holes, outside the component area. Their height has to match the cooler hardware: too tall and the cold plate never reaches the die.
Watch for	Standoff height, thread pitch and screw length work as a system. The screws must engage enough thread without bottoming out. Real cards mix threaded bosses, spring screws and insulating washers.

Check yourself: What sets how hard the cooler presses on the die? **Answer: The standoff height and the screw travel**

Reference: TI · Heat sinks and thermal interfaces

16 Bridge the gaps with pads

To place

COOLING · TP1 · TP2 · TP3 · TP4 · TP5

Place TP1, TP2, TP3, TP4 and TP5. Place TP1 through TP4 on the memory packages and TP5 over the power stages. Pads fill the space between a component and the cold plate.

What it does	A thermal pad is a soft, compressible gap filler. It conducts heat across the imperfect surfaces between a package and the cooler — less well than metal, far better than air.
Where it goes	Each pad covers the part it serves. Pad thickness must match the gap at that part: over-compressing one area can lift the cold plate away from the die.
Watch for	Pads are rated by conductivity, hardness and thickness. A thick soft pad conforms easily but adds resistance; a thin pad needs flatter surfaces. Pads do not replace paste on the die.

Check yourself: What happens if a pad is much thicker than its gap? **Answer: It holds the cooler away from the die**

Reference: TI · Heat sinks and thermal interfaces

17 Apply thermal paste

To place

COOLING · TIM1

Place TIM1. Place TIM1 on the GPU package. The goal is a thin, even film, not a thick layer.

What it does	Thermal paste fills the microscopic valleys between the die and the cold plate. Metal-to-metal contact still carries most of the heat; the paste displaces air pockets that would otherwise insulate.
Where it goes	TIM1 covers the die area at the centre of U1. The mounting pressure spreads it, so the finished layer should be thin enough to look almost transparent.
Watch for	More paste is not better: excess can pump out, trap air or spread onto nearby components. Paste performance depends on pressure and surface flatness as much as on its conductivity rating.

Check yourself: What is the paste actually replacing? **Answer: Air trapped in the surface gaps**

Reference: TI · Heat sinks and thermal interfaces

18 Create a path for heat

To place

COOLING · HS1

Place HS1. Add HS1 over the prepared thermal interface. A cutaway overlay keeps the board components visible while showing the cooling footprint.

What it does	The heatsink spreads heat from the package through a thermal interface into a larger surface exposed to air.
Where it goes	The suggested envelope spans the GPU and memory area and covers the pads and paste underneath. Real mounting, contact pressure, flatness and component clearances require mechanical design.
Watch for	This is a geometric teaching envelope. No junction temperature, thermal resistance or cooling capacity is simulated.

Check yourself: What role does the thermal interface material serve? **Answer: Improve the thermal contact between surfaces**

Reference: TI · Heat sinks and thermal interfaces

19 Move air through the cooler

To place

COOLING · FAN1 · FAN2 · FAN3

Place FAN1, FAN2 and FAN3. Finish the cooling stack with FAN1, FAN2 and FAN3 — one fan for each section of the fin stack.

What it does	A fan converts electrical power into airflow and static pressure. The heatsink needs enough pressure to push air between closely spaced fins, and the fan's curve meets the fin stack's resistance at an operating point.
Where it goes	The three fans sit side by side across the heatsink, each covering its own section of fins. Real cards add a shroud so air passes through the fins instead of escaping around them.
Watch for	Bearing type, blade geometry, PWM control and zero-RPM modes change the noise and thermal balance. More fans do not scale cooling linearly: the same air cannot be pushed through the same fins twice.

Check yourself: What limits how much heat a fan can move out of the fins? **Answer: How much air actually passes through the fin stack**

Reference: Noctua · PWM fan specifications

20 Clamp the stack with a backplate

To place

ASSEMBLY · BP1 · H5 · H6 · H7 · H8

Place BP1, H5, H6, H7 and H8. Add BP1 behind the board and place the four spring screws H5 through H8.

What it does	The backplate stiffens the PCB and spreads the cooler's clamping load, so the board does not bow under four screws. Spring screws keep that force roughly constant as the assembly expands.
Where it goes	BP1 follows the board outline on its reverse side; H5–H8 pass through it into the cooler mounting points around the GPU. Tighten evenly rather than to maximum torque.
Watch for	Warpage changes die contact and can crack BGA joints over time. Real assemblies specify a torque value and a tightening order; this bench only shows the parts and their roles.

Check yourself: Why tighten cooler screws in a cross pattern? **Answer: To load the board evenly**

Reference: TI · Heat sinks and thermal interfaces

21 Fit the PCIe bracket

To place

INTERFACES · BR1 · H9 · H10

Place BR1, H9 and H10. Place BR1 and its two screws. The bracket anchors the card to the chassis and carries the display connectors.

What it does	The bracket is structural. It transfers the card's weight into the case, aligns the edge connector with its slot and gives the display ports a rigid face.
Where it goes	BR1 sits at the board's I/O edge with J3's ports through its openings. H9 and H10 fix it to the board; the tab at the top screws into the chassis.
Watch for	Bracket geometry follows the case and PCIe mechanical specifications. A card that is too tall, too long or too thick can foul drives, cables or the side panel.

Check yourself: What does the bracket carry besides the display ports? **Answer: The card's mechanical load into the case**

Reference: NVIDIA · GPU architecture fundamentals

22 Connect the power cable

To place

POWER · CB1

Place CB1. Add CB1, the cable that feeds J2 from the power supply, and check that it seats fully.

What it does	The cable carries the card's input current. Contact resistance at a partly seated connector turns into heat, which is why seating, keying and bend radius matter.
Where it goes	CB1 runs from J2 toward the case's cable channel with enough slack to reach without pulling on the plug. Avoid a tight bend right at the connector.
Watch for	Connector standards fix pin count, keying and current per pin. This model does not assign a 12V-2x6 or 8-pin pinout, a current rating or a safe power limit.

Check yourself: Why is a partly seated power connector a risk? **Answer: Higher contact resistance heats the pins**

Reference: TI · Multiphase processor power

Further reading

- NVIDIA · GPU architecture fundamentals
- TI · Multiphase processor power
- Micron · Graphics memory
- NVIDIA · GA102 architecture whitepaper
- Winbond · Code-storage flash memory
- SiTime · Clock-generation reference example
- Noctua · PWM fan specifications
- TI · High-speed PCB layout guidance
- TI · Heat sinks and thermal interfaces

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